

Chapter 9 Exercises

SOLUTIONS TO RICK DURRETT'S
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 9 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter9_Knowledge`. The cited knowledge IDs refer to entries in `chapter9_knowledge_pieces.jsonl`.

Section 9.2 Exercises

Question. The local Chapter 9 textbook PDF has no separately numbered exercises inside Section 9.2, “Heat Equation.” The items below solve the natural verification problems in that section: the martingale calculation, uniqueness, recovery of the initial condition, and the kernel-smoothing detail left to the reader in Theorem 9.2.5.

Knowledge used. [durrett_ch9_heat_backward_martingale](#);
[durrett_ch9_heat_equation_solution](#); [durrett_ch9_heat_kernel_smoothing](#).

Exercise 9.2.A: the backward heat martingale

Problem. Suppose $u \in C^{1,2}((0, \infty) \times \mathbb{R}^d)$ and

$$u_t = \frac{1}{2} \Delta u.$$

Show that, for fixed $t > 0$,

$$M_s = u(t - s, B_s), \quad 0 \leq s < t,$$

is a local martingale.

Solution. Apply Itô’s formula to the space-time function $F(s, x) = u(t - s, x)$. Its derivatives are

$$F_s(s, x) = -u_t(t - s, x), \quad \nabla_x F(s, x) = \nabla u(t - s, x), \quad \Delta_x F(s, x) = \Delta u(t - s, x).$$

Therefore

$$du(t - s, B_s) = \left[-u_t(t - s, B_s) + \frac{1}{2} \Delta u(t - s, B_s) \right] ds + \nabla u(t - s, B_s) \cdot dB_s.$$

The drift vanishes because $u_t = (1/2) \Delta u$. Thus

$$M_s - M_0 = \int_0^s \nabla u(t - r, B_r) \cdot dB_r.$$

The stochastic integral is a local martingale after the usual localization, for instance by stopping when

$$\int_0^s |\nabla u(t - r, B_r)|^2 dr$$

exceeds n . Hence M_s is a local martingale on $[0, t)$.

Exercise 9.2.B: uniqueness of bounded heat solutions

Problem. Let u be a bounded solution of

$$u_t = \frac{1}{2}\Delta u, \quad u(0, x) = f(x),$$

where f is bounded and continuous. Prove that

$$u(t, x) = \mathbb{E}_x f(B_t).$$

Solution. Fix $t > 0$ and start Brownian motion at $B_0 = x$. By Exercise 9.2.A, $M_s = u(t - s, B_s)$ is a local martingale for $0 \leq s < t$. Since u is bounded, the localized martingales are uniformly bounded; hence M_s is an honest bounded martingale on each interval $[0, t - \epsilon]$, and the martingale convergence theorem gives

$$M_t := \lim_{s \uparrow t} M_s$$

almost surely and in L^1 .

Because $B_s \rightarrow B_t$ almost surely and u is continuous at $(0, B_t)$ with boundary value $f(B_t)$,

$$M_t = \lim_{s \uparrow t} u(t - s, B_s) = f(B_t) \quad \text{a.s.}$$

Taking expectations in the bounded martingale gives

$$u(t, x) = M_0 = \mathbb{E}_x M_t = \mathbb{E}_x f(B_t).$$

Thus a bounded solution is unique and must be the Brownian heat semigroup applied to f .

Exercise 9.2.C: the initial condition for the Brownian formula

Problem. Let

$$v(t, x) = \mathbb{E}_x f(B_t) = \mathbb{E} f(x + B_t),$$

where f is bounded and continuous. Show that

$$v(t_n, x_n) \rightarrow f(x)$$

whenever $t_n \downarrow 0$ and $x_n \rightarrow x$.

Solution. Write $B_{t_n} \stackrel{d}{=} \sqrt{t_n}Z$, where Z is a standard d -dimensional normal vector. Then

$$v(t_n, x_n) = \mathbb{E} f(x_n + \sqrt{t_n}Z).$$

For each fixed Z , $x_n + \sqrt{t_n}Z \rightarrow x$. Since f is continuous,

$$f(x_n + \sqrt{t_n}Z) \rightarrow f(x) \quad \text{a.s.}$$

Also $|f(x_n + \sqrt{t_n}Z)| \leq \|f\|_\infty$. The bounded convergence theorem gives

$$v(t_n, x_n) \rightarrow \mathbb{E} f(x) = f(x).$$

So the Brownian formula has the required initial trace.

Exercise 9.2.D: heat-kernel smoothing and the PDE

Problem. Fill in the details in Theorem 9.2.5. For bounded continuous f , prove that

$$v(t, x) = \mathbb{E}_x f(B_t) = \int_{\mathbb{R}^d} p_t(x - y) f(y) dy, \quad p_t(z) = (2\pi t)^{-d/2} e^{-|z|^2/(2t)},$$

belongs to $C^{1,2}((0, \infty) \times \mathbb{R}^d)$ and satisfies

$$v_t = \frac{1}{2} \Delta v.$$

Solution. Fix $0 < a < b < \infty$ and a compact set $K \subset \mathbb{R}^d$. We first justify differentiating under the integral for $(t, x) \in [a, b] \times K$. For a multi-index α and an integer $m \geq 0$, the derivatives

$$\partial_t^m \partial_x^\alpha p_t(x - y)$$

are finite sums of terms of the form

$$C t^{-q} (x - y)^\beta p_t(x - y),$$

where q and β depend on m, α, d . On $t \in [a, b]$ these are bounded by an integrable Gaussian-tail majorant

$$C_{a,b,K} (1 + |y|^N) e^{-c|y|^2}$$

for suitable constants $C_{a,b,K}, N, c > 0$. Since f is bounded, the dominated convergence theorem allows us to differentiate under the integral for one time derivative and for two spatial derivatives. Thus $v \in C^{1,2}$ on $(0, \infty) \times \mathbb{R}^d$.

It remains to identify the PDE. The Gaussian kernel satisfies

$$\partial_t p_t(z) = \left(-\frac{d}{2t} + \frac{|z|^2}{2t^2} \right) p_t(z),$$

while

$$\partial_{z_i} p_t(z) = -\frac{z_i}{t} p_t(z), \quad \partial_{z_i z_i} p_t(z) = \left(\frac{z_i^2}{t^2} - \frac{1}{t} \right) p_t(z).$$

Summing over i gives

$$\Delta p_t(z) = \left(\frac{|z|^2}{t^2} - \frac{d}{t} \right) p_t(z).$$

Hence

$$\partial_t p_t(z) = \frac{1}{2} \Delta p_t(z).$$

Differentiating the convolution under the integral,

$$v_t(t, x) = \int_{\mathbb{R}^d} \partial_t p_t(x - y) f(y) dy = \frac{1}{2} \int_{\mathbb{R}^d} \Delta_x p_t(x - y) f(y) dy = \frac{1}{2} \Delta v(t, x).$$

Therefore v is a classical solution of the heat equation for $t > 0$. Together with Exercise 9.2.C, this proves existence of a bounded solution with initial value f ; Exercise 9.2.B proves uniqueness.

Section 9.3 Exercises

Question. The local Chapter 9 textbook PDF also has no separately numbered exercise block inside Section 9.3, “Inhomogeneous Heat Equation.” The solutions below spell out the section’s verification problems for the equation

$$u_t = \frac{1}{2}\Delta u + g, \quad u(0, x) = 0,$$

with bounded continuous g . For nonzero initial condition f , add the homogeneous heat solution $\mathbb{E}_x f(B_t)$ from Section 9.2.

Knowledge used. [durrett_ch9_duhamel_inhomogeneous_heat](#);
[durrett_ch9_inhomogeneous_heat_regularity](#); [durrett_ch9_heat_kernel_smoothing](#).

Exercise 9.3.A: the source-term martingale

Problem. Suppose $u \in C^{1,2}$ and

$$u_t = \frac{1}{2}\Delta u + g.$$

Show that, for fixed $t > 0$,

$$M_s = u(t - s, B_s) + \int_0^s g(B_r) dr, \quad 0 \leq s < t,$$

is a local martingale.

Solution. As in Section 9.2, apply Itô’s formula to $u(t - s, B_s)$:

$$du(t - s, B_s) = \left[-u_t(t - s, B_s) + \frac{1}{2}\Delta u(t - s, B_s) \right] ds + \nabla u(t - s, B_s) \cdot dB_s.$$

Since $u_t = (1/2)\Delta u + g$,

$$-u_t + \frac{1}{2}\Delta u = -g.$$

Hence

$$d \left(u(t - s, B_s) + \int_0^s g(B_r) dr \right) = \nabla u(t - s, B_s) \cdot dB_s.$$

After localizing the stochastic integral by its quadratic variation, the right side is a martingale. Therefore M_s is a local martingale.

Exercise 9.3.B: uniqueness and Duhamel’s formula

Problem. Assume u is bounded on $[0, T] \times \mathbb{R}^d$ for each $T < \infty$, satisfies

$$u_t = \frac{1}{2}\Delta u + g, \quad u(0, x) = 0,$$

and g is bounded. Prove that

$$u(t, x) = \mathbb{E}_x \int_0^t g(B_s) ds.$$

Solution. Fix $t > 0$ and $B_0 = x$. By Exercise 9.3.A,

$$M_s = u(t - s, B_s) + \int_0^s g(B_r) dr, \quad 0 \leq s < t,$$

is a local martingale. Since u and g are bounded on the relevant finite time interval, M_s is bounded by a deterministic constant, so the localized martingale is uniformly integrable and M_s is an honest martingale.

Letting $s \uparrow t$, continuity at time 0 gives

$$\lim_{s \uparrow t} u(t - s, B_s) = u(0, B_t) = 0,$$

and bounded convergence gives

$$M_t = \int_0^t g(B_r) dr.$$

Taking expectations,

$$u(t, x) = M_0 = \mathbb{E}_x M_t = \mathbb{E}_x \int_0^t g(B_s) ds.$$

This proves uniqueness among bounded solutions and identifies the solution as Duhamel's formula.

Exercise 9.3.C: smooth formula implies the PDE

Problem. Let

$$v(t, x) = \mathbb{E}_x \int_0^t g(B_s) ds.$$

Assume $v \in C^{1,2}((0, \infty) \times \mathbb{R}^d)$. Show that

$$v_t = \frac{1}{2} \Delta v + g.$$

Solution. Use the Markov property to write, for $h > 0$,

$$v(t + h, x) = \mathbb{E}_x \int_0^{t+h} g(B_s) ds = \mathbb{E}_x \int_0^h g(B_s) ds + \mathbb{E}_x v(t, B_h).$$

Subtract $v(t, x)$:

$$v(t + h, x) - v(t, x) = \mathbb{E}_x \{v(t, B_h) - v(t, x)\} + \mathbb{E}_x \int_0^h g(B_s) ds.$$

Because v is $C^{1,2}$ in space, Taylor's formula at x gives

$$\mathbb{E}_x \{v(t, B_h) - v(t, x)\} = \frac{1}{2} \Delta v(t, x) h + o(h),$$

using $\mathbb{E}_x(B_h^i - x_i) = 0$, $\mathbb{E}_x[(B_h^i - x_i)(B_h^j - x_j)] = h\delta_{ij}$. Since g is continuous and $B_s \rightarrow x$ in probability as $s \downarrow 0$,

$$\frac{1}{h} \mathbb{E}_x \int_0^h g(B_s) ds \rightarrow g(x).$$

Divide by h and let $h \downarrow 0$:

$$v_t(t, x) = \frac{1}{2} \Delta v(t, x) + g(x).$$

Exercise 9.3.D: initial condition

Problem. Show that

$$v(t, x) = \mathbb{E}_x \int_0^t g(B_s) ds$$

satisfies $v(t, x) \rightarrow 0$ as $t \downarrow 0$, uniformly in x , when g is bounded.

Solution. If $\|g\|_\infty \leq M$, then

$$|v(t, x)| \leq \mathbb{E}_x \int_0^t |g(B_s)| ds \leq Mt.$$

Therefore $\sup_x |v(t, x)| \leq Mt \rightarrow 0$. This is exactly the initial condition $v(0, x) = 0$.

Exercise 9.3.E: kernel form and regularity

Problem. Write the Duhamel solution in heat-kernel form and explain why Hölder continuity of g is a natural sufficient condition for $v \in C^{1,2}$.

Solution. By Fubini's theorem and the Brownian transition density,

$$v(t, x) = \int_0^t \mathbb{E}_x g(B_s) ds = \int_0^t \int_{\mathbb{R}^d} p_s(x - y) g(y) dy ds.$$

Equivalently,

$$v(t, x) = \int_{\mathbb{R}^d} G_t(x, y) g(y) dy, \quad G_t(x, y) = \int_0^t p_s(x - y) ds.$$

Formally differentiating under the integral is delicate near $s = 0$, because derivatives of p_s become singular as $s \downarrow 0$ and $y \rightarrow x$. The cancellation supplied by

$$g(y) = g(x) + O(|y - x|^\alpha)$$

when g is Hölder continuous offsets this singularity. More concretely, split

$$P_s g(x) - g(x) = \int_{\mathbb{R}^d} p_s(z) \{g(x + z) - g(x)\} dz.$$

If g is Hölder with exponent $\alpha > 0$, then

$$|P_s g(x) - g(x)| \leq C \mathbb{E} |B_s|^\alpha = C' s^{\alpha/2}.$$

This is the small-time gain that makes the singular kernel estimates integrable in the parabolic regularity argument. The standard Schauder estimate for the heat equation then gives

$$v \in C^{1,2}((0, \infty) \times \mathbb{R}^d),$$

and Exercise 9.3.C proves that this classical function solves

$$v_t = \frac{1}{2} \Delta v + g, \quad v(0, x) = 0.$$

Thus Hölder continuity is not just a cosmetic assumption; it is the condition that lets the source term smooth into a classical solution.

Section 9.4 Exercises

Question. The local Chapter 9 textbook PDF has no separately numbered exercise block inside Section 9.4, “Feynman-Kac Formula.” The solutions below work through the section’s verification problems for

$$u_t = \frac{1}{2}\Delta u + c(x)u, \quad u(0, x) = f(x),$$

where c and f are bounded and continuous.

Knowledge used. [durrett_ch9_feynman_kac_forward](#); [durrett_ch9_feynman_kac_smoothness](#); [durrett_ch9_heat_equation_solution](#); [durrett_ch9_duhamel_inhomogeneous_heat](#).

Exercise 9.4.A: the weighted martingale

Problem. Let

$$A_s = \int_0^s c(B_r) dr.$$

Suppose $u \in C^{1,2}$ and

$$u_t = \frac{1}{2}\Delta u + c(x)u.$$

Show that, for fixed $t > 0$,

$$M_s = u(t - s, B_s)e^{A_s}, \quad 0 \leq s < t,$$

is a local martingale.

Solution. The process A_s has finite variation, so the ordinary product rule and Itô’s formula give

$$d\{u(t - s, B_s)e^{A_s}\} = e^{A_s} du(t - s, B_s) + u(t - s, B_s)e^{A_s} dA_s.$$

There is no quadratic covariation term because e^{A_s} has finite variation. Also

$$dA_s = c(B_s) ds$$

and

$$du(t - s, B_s) = \left[-u_t(t - s, B_s) + \frac{1}{2}\Delta u(t - s, B_s) \right] ds + \nabla u(t - s, B_s) \cdot dB_s.$$

Therefore

$$dM_s = e^{A_s} \left[-u_t + \frac{1}{2}\Delta u + cu \right] (t - s, B_s) ds + e^{A_s} \nabla u(t - s, B_s) \cdot dB_s.$$

Since $u_t = (1/2)\Delta u + cu$, the drift vanishes. After localizing the stochastic integral by its quadratic variation,

$$M_s - M_0 = \int_0^s e^{A_r} \nabla u(t - r, B_r) \cdot dB_r$$

is a local martingale. Hence M_s is a local martingale.

Exercise 9.4.B: uniqueness and the Feynman-Kac formula

Problem. Assume u is bounded on $[0, T] \times \mathbb{R}^d$ for each $T < \infty$ and satisfies

$$u_t = \frac{1}{2} \Delta u + c(x)u, \quad u(0, x) = f(x),$$

with c bounded. Prove that

$$u(t, x) = \mathbb{E}_x \left[f(B_t) \exp \left(\int_0^t c(B_s) ds \right) \right].$$

Solution. Fix $t > 0$ and start Brownian motion at $B_0 = x$. By Exercise 9.4.A,

$$M_s = u(t - s, B_s) \exp \left(\int_0^s c(B_r) dr \right)$$

is a local martingale on $[0, t)$. If $|c| \leq C$ and u is bounded on $[0, t] \times \mathbb{R}^d$, then

$$|M_s| \leq e^{Ct} \sup_{0 \leq r \leq t, y \in \mathbb{R}^d} |u(r, y)|.$$

Thus M_s is bounded and therefore an honest uniformly integrable martingale.

Let $s \uparrow t$. Since $B_s \rightarrow B_t$ almost surely and u is continuous at the initial boundary,

$$\lim_{s \uparrow t} u(t - s, B_s) = u(0, B_t) = f(B_t).$$

Also $\int_0^s c(B_r) dr \rightarrow \int_0^t c(B_r) dr$. Hence

$$M_t = f(B_t) \exp \left(\int_0^t c(B_r) dr \right).$$

Taking expectations in the martingale identity $M_0 = \mathbb{E}_x M_t$ gives the claimed formula. In particular, this proves uniqueness among bounded solutions.

Exercise 9.4.C: a smooth Feynman-Kac formula solves the PDE

Problem. Let

$$v(t, x) = \mathbb{E}_x \left[f(B_t) \exp \left(\int_0^t c(B_s) ds \right) \right].$$

Assume $v \in C^{1,2}((0, \infty) \times \mathbb{R}^d)$. Show that

$$v_t = \frac{1}{2} \Delta v + c(x)v.$$

Solution. The Markov property gives, for $h > 0$,

$$v(t + h, x) = \mathbb{E}_x \left[e^{\int_0^h c(B_s) ds} v(t, B_h) \right].$$

Subtract $v(t, x)$ and add-subtract $v(t, B_h)$:

$$v(t + h, x) - v(t, x) = \mathbb{E}_x \{ v(t, B_h) - v(t, x) \} + \mathbb{E}_x \left[v(t, B_h) \left(e^{\int_0^h c(B_s) ds} - 1 \right) \right].$$

Since v is $C^{1,2}$ in space,

$$\mathbb{E}_x\{v(t, B_h) - v(t, x)\} = \frac{1}{2}\Delta v(t, x)h + o(h).$$

For the second term, boundedness of c gives

$$e^{\int_0^h c(B_s) ds} - 1 = \int_0^h c(B_r) \exp\left(\int_r^h c(B_s) ds\right) dr.$$

As $h \downarrow 0$, $B_r \rightarrow x$ in probability uniformly on small time intervals and $v(t, B_h) \rightarrow v(t, x)$ in probability. The integrand is bounded on compact time intervals, so dominated convergence yields

$$\frac{1}{h}\mathbb{E}_x\left[v(t, B_h)\left(e^{\int_0^h c(B_s) ds} - 1\right)\right] \rightarrow c(x)v(t, x).$$

Divide by h and let $h \downarrow 0$ to obtain

$$v_t(t, x) = \frac{1}{2}\Delta v(t, x) + c(x)v(t, x).$$

Exercise 9.4.D: initial condition

Problem. Show that the Feynman-Kac formula satisfies the initial condition:

$$v(t_n, x_n) \rightarrow f(x)$$

whenever $t_n \downarrow 0$ and $x_n \rightarrow x$.

Solution. Write

$$A_t = \int_0^t c(B_s) ds.$$

If $|c| \leq C$, then $|A_t| \leq Ct$, so $e^{A_t} \rightarrow 1$ uniformly as $t \downarrow 0$. Therefore

$$|v(t, x) - \mathbb{E}_x f(B_t)| \leq \|f\|_\infty \mathbb{E}_x |e^{A_t} - 1| \leq \|f\|_\infty (e^{Ct} - 1) \rightarrow 0.$$

By the Section 9.2 initial-condition result,

$$\mathbb{E}_{x_n} f(B_{t_n}) \rightarrow f(x).$$

Combining the two estimates gives

$$v(t_n, x_n) \rightarrow f(x).$$

Exercise 9.4.E: reducing smoothness to Sections 9.2 and 9.3

Problem. Assume f and c are Holder continuous and bounded. Explain the identity used in Durrett's proof of Theorem 9.4.5 and why it reduces smoothness of the Feynman-Kac formula to the heat and inhomogeneous heat results.

Solution. Let

$$A_t = \int_0^t c(B_r) dr.$$

Since A_t is continuous and of finite variation,

$$e^{A_t} - 1 = \int_0^t c(B_s) e^{A_t - A_s} ds = \int_0^t c(B_s) \exp\left(\int_s^t c(B_r) dr\right) ds.$$

Multiplying by $f(B_t)$ and taking expectations,

$$v(t, x) = \mathbb{E}_x f(B_t) + \int_0^t \mathbb{E}_x \left[c(B_s) \exp \left(\int_s^t c(B_r) dr \right) f(B_t) \right] ds.$$

Conditioning on $\mathcal{F}_s$ and using the Markov property turns the second integrand into

$$\mathbb{E}_x [c(B_s) v(t-s, B_s)].$$

Thus

$$v(t, x) = v_1(t, x) + v_2(t, x),$$

where

$$v_1(t, x) = \mathbb{E}_x f(B_t), \quad v_2(t, x) = \int_0^t \mathbb{E}_x \{c(B_s) v(t-s, B_s)\} ds.$$

The first term is the homogeneous heat solution from Section 9.2, so $v_1 \in C^{1,2}$ for $t > 0$. The second term has the form of the inhomogeneous heat solution from Section 9.3 with source

$$g(r, x) = c(x) v(r, x).$$

Boundedness of c and f makes v locally bounded in time. Holder continuity of f gives local Holder control of v_1 in x , while the inhomogeneous heat estimates give the same kind of local Holder control for v_2 . Since c is Holder, the product $g(r, x) = c(x) v(r, x)$ is locally Holder in the spatial variable. Section 9.3's regularity result then gives $v_2 \in C^{1,2}$, and hence

$$v = v_1 + v_2 \in C^{1,2}.$$

Exercise 9.4.C then shows that this smooth Feynman-Kac formula solves the PDE.

Section 9.5 Exercises

Knowledge used. `durrett_ch9_dirichlet_problem_representation;`
`durrett_ch9_mean_value_harmonicity;` `durrett_ch9_interior_smoothness_harmonic;`
`durrett_ch9_regular_boundary_point;` `durrett_ch9_boundary_continuity_regular_points;`
`durrett_ch9_cone_condition.`

Exercise 9.5.1

Question. Suppose h is bounded and harmonic on $\mathbb{R}^d$. Prove that h is constant.

Solution. Let B_t be Brownian motion started at x . Since h is harmonic, $h(B_{t \wedge S_R})$ is a bounded martingale, where $S_R = \inf\{t : |B_t - x| = R\}$. Letting $R \rightarrow \infty$ and using bounded convergence gives

$$h(x) = \mathbb{E} h(x + B_t), \quad t > 0.$$

Thus, for any $x, y \in \mathbb{R}^d$,

$$h(x) - h(y) = \int_{\mathbb{R}^d} h(z) \{p_t(z - x) - p_t(z - y)\} dz.$$

If $\|h\|_\infty \leq M$, then

$$|h(x) - h(y)| \leq M \int_{\mathbb{R}^d} |p_t(z - x) - p_t(z - y)| dz.$$

The two Gaussian densities have the same covariance matrix tI and means separated by the fixed vector $x - y$. As $t \rightarrow \infty$, their total variation distance tends to 0. Hence $|h(x) - h(y)| = 0$. Since x and y were arbitrary, h is constant.

Exercise 9.5.2

Question. Suppose $f \in C^2$ is superharmonic, takes values in $[0, \infty]$, and is continuous. Show that in dimensions 1 and 2, f must be constant. Give an example of the form $f(x) = g(|x|)$ in $d \geq 3$.

Solution. Use the convention that superharmonic means $\Delta f \leq 0$. Itô's formula gives

$$f(B_t) - f(B_0) = \int_0^t \nabla f(B_s) \cdot dB_s + \frac{1}{2} \int_0^t \Delta f(B_s) ds.$$

After localization, $f(B_t)$ is therefore a nonnegative local supermartingale, hence a supermartingale.

Let $x, y \in \mathbb{R}^d$ and let

$$T_\varepsilon = \inf\{t : |B_t - y| \leq \varepsilon\}$$

for Brownian motion started at x . In dimensions 1 and 2, Brownian motion is recurrent, so $T_\varepsilon < \infty$ almost surely. Optional stopping at $T_\varepsilon \wedge n$ and nonnegativity give

$$f(x) \geq \mathbb{E}_x f(B_{T_\varepsilon});$$

then let $n \rightarrow \infty$. Since B_{T_ε} lies on the sphere of radius ε about y , continuity implies

$$\lim_{\varepsilon \downarrow 0} \mathbb{E}_x f(B_{T_\varepsilon}) = f(y).$$

Thus $f(x) \geq f(y)$. Reversing x and y gives $f(y) \geq f(x)$, so f is constant.

For $d \geq 3$, a smooth nonconstant radial example is

$$f(x) = g(|x|) = (1 + |x|^2)^{-(d-2)/2}.$$

Let $\alpha = (d-2)/2$ and $r = |x|$. For $g(r) = (1 + r^2)^{-\alpha}$,

$$g''(r) + \frac{d-1}{r} g'(r) = 2\alpha(1 + r^2)^{-\alpha-2} \{-d + (2\alpha + 2 - d)r^2\}.$$

Since $2\alpha + 2 - d = 0$, this becomes

$$\Delta f(x) = -d(d-2)(1 + |x|^2)^{-(d+2)/2} \leq 0.$$

So f is nonnegative, smooth, radial, superharmonic, and nonconstant.

Exercise 9.5.3

Question. Let $\tau = \inf\{t : B_t \notin G\}$. Suppose G is connected and open, all boundary points are regular, and $\mathbb{P}_x(\tau < \infty) < 1$ in G . Show that all bounded solutions of the Dirichlet problem with boundary value 0 on ∂G have the form

$$u(x) = c\mathbb{P}_x(\tau = \infty).$$

Solution. Put

$$q(x) = \mathbb{P}_x(\tau = \infty).$$

By the strong Markov property, q is harmonic in G . Since every boundary point is regular, $q(x) \rightarrow 0$ as x approaches ∂G . Thus q itself is a bounded solution of the zero-boundary Dirichlet problem. The assumption $\mathbb{P}_x(\tau < \infty) < 1$ says $q(x) > 0$ in G .

Let u be another bounded solution with zero boundary data and let $M = \|u\|_\infty$. Stopping $u(B_t)$ at τ and using the regular zero boundary condition gives

$$u(x) = \mathbb{E}_x\{u(B_t); \tau > t\}.$$

Therefore

$$|u(x)| \leq M\mathbb{P}_x(\tau > t).$$

Letting $t \rightarrow \infty$ gives

$$|u(x)| \leq Mq(x).$$

Hence $h(x) = u(x)/q(x)$ is bounded in G .

Now use the Doob q -transform, i.e. Brownian motion conditioned never to leave G :

$$\mathbb{P}_x^q(A) = \frac{1}{q(x)} \mathbb{E}_x\{\mathbf{1}_A q(B_t); t < \tau\}, \quad A \in \mathcal{F}_t.$$

The ratio $h = u/q$ is harmonic for this conditioned process. Since G is connected and all finite boundary points are regular, the only remaining boundary point for the conditioned process is the ideal point “at infinity.” The bounded martingale $h(B_t)$ therefore has a constant terminal value under the conditioned law. Calling this constant c , we get

$$h(x) = \mathbb{E}_x^q h(B_t) = c,$$

and hence

$$u(x) = cq(x) = c\mathbb{P}_x(\tau = \infty).$$

Equivalently, this is the maximum-principle content of the hint: adding a large multiple of q to $\pm u$ produces nonnegative zero-boundary harmonic functions, and the only possible freedom is the amount of mass assigned to escape through infinity.

Exercise 9.5.4

Question. Let $\tau = \inf\{t : B_t \notin G\}$ and consider the Dirichlet problem in a connected open set G in $d \geq 3$, where all boundary points are regular and $\mathbb{P}_x(\tau < \infty) < 1$ in G . The extracted statement again has boundary condition $f \equiv 0$. Conclude that all bounded solutions have the form

$$u(x) = c\mathbb{P}_x(\tau = \infty).$$

Solution. For the zero-boundary statement, this is exactly Exercise 9.5.3. The dimension assumption $d \geq 3$ is the natural transient setting in which escape to infinity may have positive probability.

The usual general form is worth recording. If the boundary value is a bounded continuous function f , one particular solution is

$$v(x) = \mathbb{E}_x\{f(B_\tau); \tau < \infty\}.$$

If u is any other bounded solution with the same boundary data, then $w = u - v$ is a bounded solution with zero boundary data. By Exercise 9.5.3,

$$w(x) = c\mathbb{P}_x(\tau = \infty).$$

Hence all bounded solutions with boundary value f are

$$u(x) = \mathbb{E}_x\{f(B_\tau); \tau < \infty\} + c\mathbb{P}_x(\tau = \infty).$$

When $f \equiv 0$, this reduces to the displayed formula in the question.

Exercise 9.5.5

Question. Define a flat cone $\bar{V}(y, v, a)$ to be the intersection of $V(y, v, a)$ with a $(d-1)$ -dimensional hyperplane that contains the line $\{y + \theta v : \theta \in \mathbb{R}\}$. Show that Theorem 9.5.10 remains true if “cone” is replaced by “flat cone.”

Solution. By translating and rotating, take $y = 0$ and take the hyperplane to be

$$H = \{x \in \mathbb{R}^d : x_d = 0\}.$$

Write Brownian motion as $B_t = (X_t, Y_t)$, where X_t is $(d-1)$ -dimensional Brownian motion in H and Y_t is an independent one-dimensional Brownian motion perpendicular to H .

The zero set of Y_t accumulates at 0 almost surely. At those times Brownian motion lies in the hyperplane H . The trace of B_t on H , parametrized by the local time of Y_t at 0, is a rotationally symmetric Cauchy process in H . Starting from 0, this trace process enters every relatively open cone in H immediately with probability one. Therefore, if G^c contains a flat cone with vertex y , then Brownian motion started at y hits G^c at arbitrarily small positive times almost surely.

Thus

$$\mathbb{P}_y(\tau_G = 0) = 1,$$

so y is a regular boundary point. This is exactly the conclusion of Theorem 9.5.10 with “cone” replaced by “flat cone.”

Section 9.6 Exercises

Question. The local Chapter 9 PDF has no separately numbered exercise block in Section 9.6, “Green’s Functions and Potential Kernels.” The items below solve the section’s natural verification problems: occupation times of balls, the potential-kernel computations in dimensions $d \geq 3$, $d = 2$, and $d = 1$, and the normalization $(1/2)\Delta G = -\delta_0$.

Knowledge used. [durrett_ch9_newtonian_potential_kernel](#);
[durrett_ch9_green_function_killed_domain](#); [durrett_ch9_radial_harmonic_functions](#).

Exercise 9.6.A: occupation time of a ball

Problem. Let $D = B(0, r)$. Show that

$$\mathbb{P}_x \left(\int_0^\infty \mathbf{1}_D(B_t) dt = \infty \right) = 1 \quad \text{in } d \leq 2,$$

while

$$\mathbb{E}_x \int_0^\infty \mathbf{1}_D(B_t) dt < \infty \quad \text{in } d \geq 3.$$

Solution. For $d \leq 2$, Brownian motion is recurrent. Thus it enters D infinitely often almost surely. Let $G = B(0, 2r)$ and define successive entrance and exit times

$$S_k = \inf\{t > T_{k-1} : B_t \in D\}, \quad T_k = \inf\{t > S_k : B_t \notin G\},$$

with $T_0 = 0$. By the strong Markov property, the occupation increments

$$X_k = \int_{S_k}^{T_k} \mathbf{1}_D(B_t) dt$$

are independent after conditioning on the entrance point, and each has a positive mean bounded away from 0 because Brownian motion started in the compact ball D spends positive expected time in D before leaving G . Hence, by the strong law applied along visits,

$$\sum_{k=1}^{\infty} X_k = \infty \quad \text{a.s.}$$

Therefore the total occupation time of D is infinite almost surely.

For $d \geq 3$, use Fubini and the transition density

$$p_t(x, y) = (2\pi t)^{-d/2} e^{-|x-y|^2/(2t)}.$$

For nonnegative f ,

$$\mathbb{E}_x \int_0^{\infty} f(B_t) dt = \int_{\mathbb{R}^d} \left(\int_0^{\infty} p_t(x, y) dt \right) f(y) dy.$$

The inner integral is finite for $x \neq y$ when $d \geq 3$ and equals a constant times $|x-y|^{2-d}$, as computed in Exercise 9.6.B. Taking $f = \mathbf{1}_D$, the singularity is locally integrable:

$$\int_{B(0,r)} |x-y|^{2-d} dy < \infty,$$

because near $y = x$ the polar-coordinate integrand is $\rho^{2-d}\rho^{d-1} = \rho$. Thus

$$\mathbb{E}_x \int_0^{\infty} \mathbf{1}_D(B_t) dt < \infty.$$

Exercise 9.6.B: the potential kernel in $d \geq 3$

Problem. Compute

$$G(x, y) = \int_0^{\infty} p_t(x, y) dt$$

for $d \geq 3$.

Solution. Let $a = |x - y|$. For $a > 0$,

$$G(x, y) = \int_0^{\infty} (2\pi t)^{-d/2} e^{-a^2/(2t)} dt.$$

Set $s = a^2/(2t)$, so $t = a^2/(2s)$ and

$$dt = -\frac{a^2}{2s^2} ds.$$

Then

$$\begin{aligned} G(x, y) &= \int_{\infty}^0 \left(2\pi \frac{a^2}{2s} \right)^{-d/2} e^{-s} \left(-\frac{a^2}{2s^2} \right) ds \\ &= \frac{a^{2-d}}{2\pi^{d/2}} \int_0^{\infty} s^{d/2-2} e^{-s} ds \\ &= \frac{\Gamma(d/2 - 1)}{2\pi^{d/2}} |x - y|^{2-d}. \end{aligned}$$

The gamma integral is finite exactly when $d/2 - 1 > 0$, i.e. $d \geq 3$.

Exercise 9.6.C: the modified kernel in $d = 1$ **Problem.** In dimension 1, define

$$G(x, y) = \int_0^\infty \{p_t(x, y) - p_t(0, 0)\} dt.$$

Show that

$$G(x, y) = -|x - y|.$$

Solution. Let $a = |x - y|$. Then

$$G(x, y) = \frac{1}{\sqrt{2\pi}} \int_0^\infty \left(e^{-a^2/(2t)} - 1 \right) t^{-1/2} dt.$$

With $u = a^2/(2t)$,

$$G(x, y) = -\frac{a}{2\sqrt{\pi}} \int_0^\infty (1 - e^{-u}) u^{-3/2} du.$$

Use

$$1 - e^{-u} = \int_0^u e^{-s} ds$$

and Fubini:

$$\begin{aligned} \int_0^\infty (1 - e^{-u}) u^{-3/2} du &= \int_0^\infty e^{-s} \int_s^\infty u^{-3/2} du ds \\ &= 2 \int_0^\infty e^{-s} s^{-1/2} ds = 2\sqrt{\pi}. \end{aligned}$$

Therefore

$$G(x, y) = -a = -|x - y|.$$

Exercise 9.6.D: the modified kernel in $d = 2$ **Problem.** In dimension 2, define

$$G(x, y) = \int_0^\infty \{p_t(x, y) - p_t(0, e_1)\} dt, \quad e_1 = (1, 0).$$

Show that

$$G(x, y) = -\frac{1}{\pi} \log |x - y|.$$

Solution. Let $a = |x - y|$. Since

$$p_t(x, y) = \frac{1}{2\pi t} e^{-a^2/(2t)}, \quad p_t(0, e_1) = \frac{1}{2\pi t} e^{-1/(2t)},$$

we have

$$G(x, y) = \frac{1}{2\pi} \int_0^\infty \left(e^{-a^2/(2t)} - e^{-1/(2t)} \right) t^{-1} dt.$$

Set $u = 1/(2t)$. Then $t^{-1} dt = -u^{-1} du$, and

$$G(x, y) = \frac{1}{2\pi} \int_0^\infty \frac{e^{-a^2 u} - e^{-u}}{u} du.$$

The standard Frullani integral gives, for $a > 0$,

$$\int_0^\infty \frac{e^{-a^2 u} - e^{-u}}{u} du = \log \frac{1}{a^2} = -2 \log a.$$

Hence

$$G(x, y) = -\frac{1}{\pi} \log |x - y|.$$

Exercise 9.6.E: distributional normalization

Problem. Verify that the constants in the potential kernels are chosen so that

$$\frac{1}{2} \Delta_x G(x, 0) = -\delta_0$$

in the distributional sense.

Solution. Away from 0, each kernel is a constant multiple of the radial harmonic function from Section 9.1:

$$|x|^{2-d} \quad (d \geq 3), \quad \log |x| \quad (d = 2), \quad |x| \quad (d = 1).$$

So $\Delta G(x, 0) = 0$ for $x \neq 0$. It remains to check the mass at the origin.

For $d = 1$, $G(x, 0) = -|x|$. Since $(|x|)' = \text{sgn}(x)$ and $(|x|)'' = 2\delta_0$,

$$\frac{1}{2} G'' = -\delta_0.$$

For $d \geq 3$, write

$$G(x, 0) = C_d |x|^{2-d}, \quad C_d = \frac{\Gamma(d/2 - 1)}{2\pi^{d/2}}.$$

Let $\varphi \in C_c^\infty(\mathbb{R}^d)$. Since G is harmonic off 0, Green's identity on $\{|x| > \varepsilon\}$ gives

$$\int_{|x| > \varepsilon} G(x, 0) \Delta \varphi(x) dx = \int_{|x| = \varepsilon} \{G \partial_n \varphi - \varphi \partial_n G\} dS,$$

where n is the inward normal for the removed ball. The term $G \partial_n \varphi$ is $O(\varepsilon)$ and vanishes. Also

$$\partial_r G = C_d(2 - d)\varepsilon^{1-d}.$$

With the inward normal convention on the inner boundary, $\partial_n G = -\partial_r G = C_d(d - 2)\varepsilon^{1-d}$. Hence

$$\int_{\mathbb{R}^d} G \Delta \varphi = -C_d(d - 2)\omega_d \varphi(0),$$

where $\omega_d = 2\pi^{d/2}/\Gamma(d/2)$ is the surface area of the unit sphere. Using

$$\Gamma(d/2) = (d/2 - 1)\Gamma(d/2 - 1),$$

we get $C_d(d - 2)\omega_d = 2$. Therefore

$$\int_{\mathbb{R}^d} \frac{1}{2} G \Delta \varphi = -\varphi(0),$$

which is exactly $(1/2)\Delta G = -\delta_0$.

For $d = 2$, $G(x, 0) = -(1/\pi) \log |x|$. The same boundary calculation uses $\partial_r G = -(1/\pi)r^{-1}$ and the circle length $2\pi r$, giving

$$\int_{\mathbb{R}^2} G \Delta \varphi = -2\varphi(0), \quad \int_{\mathbb{R}^2} \frac{1}{2} G \Delta \varphi = -\varphi(0).$$

Thus the normalization is correct in every dimension.

Section 9.7 Exercises

Knowledge used. [durrett_ch9_poisson_equation_representation](#);
[durrett_ch9_expected_exit_time_poisson](#); [durrett_ch9_poisson_boundary_regular](#).

Exercise 9.7.1

Question. Let G be a bounded open set and let

$$\tau = \inf\{t : B_t \notin G\}.$$

What equation should we solve to find

$$w(x) = \mathbb{E}_x \tau^2?$$

Solution. First recall the Section 9.7 result for the first moment. If

$$v(x) = \mathbb{E}_x \tau,$$

then v solves

$$\frac{1}{2}\Delta v = -1 \quad \text{in } G, \quad v = 0 \quad \text{on } \partial G$$

at regular boundary points.

For the second moment, use the Markov property at a small deterministic time t , or equivalently Dynkin's formula. On the event $\{t < \tau\}$,

$$\tau = t + \tau \circ \theta_t,$$

so the general moment recursion is

$$\frac{1}{2}\Delta m_n = -n m_{n-1}, \quad m_n = 0 \quad \text{on } \partial G,$$

where $m_n(x) = \mathbb{E}_x \tau^n$ and $m_0 \equiv 1$. Therefore

$$w(x) = m_2(x) = \mathbb{E}_x \tau^2$$

should be found by solving

$$\boxed{\frac{1}{2}\Delta w = -2v \quad \text{in } G, \quad w = 0 \quad \text{on } \partial G,}$$

where v is the solution of the first-moment problem above.

Equivalently, eliminating v gives a fourth-order equation. Since

$$v = -\frac{1}{4}\Delta w,$$

and $(1/2)\Delta v = -1$, we get

$$\frac{1}{8}\Delta^2 w = -1, \quad \text{or} \quad \Delta^2 w = -8 \quad \text{in } G.$$

The coupled Poisson system is usually the cleaner way to state and solve the problem:

$$\frac{1}{2}\Delta v = -1, \quad v|_{\partial G} = 0, \quad \frac{1}{2}\Delta w = -2v, \quad w|_{\partial G} = 0.$$

Section 9.8 Exercises

Question. The local Chapter 9 PDF has no separately numbered exercise block in Section 9.8, “Schrodinger Equation.” The solutions below work through the section’s main verification problems: the weighted martingale, the one-dimensional obstruction, the finite-gauge condition, the Feynman-Kac representation, and the explicit exponential exit-time formula.

Knowledge used. [durrett_ch9_schrodinger_martingale](#);
[durrett_ch9_schrodinger_nonuniqueness_warning](#); [durrett_ch9_gauge_function](#);
[durrett_ch9_gauge_theorem_connected_domain](#);
[durrett_ch9_schrodinger_dirichlet_representation](#);
[durrett_ch9_schrodinger_boundary_regular](#); [durrett_ch9_interval_exponential_exit_moment](#).

Exercise 9.8.A: the Schrodinger martingale

Problem. Let G be open, let $\tau = \inf\{t : B_t \notin G\}$, and suppose $u \in C^2(G)$ satisfies

$$\frac{1}{2}\Delta u + c(x)u = 0 \quad \text{in } G.$$

Show that

$$M_t = u(B_t) \exp\left(\int_0^t c(B_s) ds\right), \quad 0 \leq t < \tau,$$

is a local martingale.

Solution. Put

$$A_t = \int_0^t c(B_s) ds.$$

Since A_t has finite variation, the product rule and Itô’s formula give, for $t < \tau$,

$$\begin{aligned} d\{u(B_t)e^{A_t}\} &= e^{A_t}\nabla u(B_t) \cdot dB_t + e^{A_t}\frac{1}{2}\Delta u(B_t) dt + u(B_t)e^{A_t}c(B_t) dt \\ &= e^{A_t}\nabla u(B_t) \cdot dB_t + e^{A_t}\left(\frac{1}{2}\Delta u + cu\right)(B_t) dt. \end{aligned}$$

The drift vanishes by the PDE. After stopping when

$$\int_0^t e^{2A_s} |\nabla u(B_s)|^2 ds$$

gets large or when B_t approaches ∂G , the stochastic integral is a martingale. Hence M_t is a local martingale on $[0, \tau)$.

Exercise 9.8.B: why boundedness alone is not enough

Problem. Solve the boundary-value problem

$$\frac{1}{2}u'' + \gamma u = 0 \quad \text{on } (-a, a), \quad u(-a) = u(a) = 1,$$

and explain why this warns us not to assert blindly that

$$u(x) = \mathbb{E}_x e^{\gamma\tau}, \quad \tau = \inf\{t : B_t \notin (-a, a)\}.$$

Solution. Let $b = \sqrt{2\gamma}$. The general solution is

$$u(x) = A \cos(bx) + B \sin(bx).$$

The boundary equations are

$$A \cos(ba) + B \sin(ba) = 1, \quad A \cos(ba) - B \sin(ba) = 1.$$

Thus $B \sin(ba) = 0$ and $A \cos(ba) = 1$. When $\cos(ba) \neq 0$, the even solution is

$$u(x) = \frac{\cos(bx)}{\cos(ba)}.$$

When $\cos(ba) = 0$, no solution with boundary value 1 exists.

The probabilistic quantity $\mathbb{E}_x e^{\gamma\tau}$ is nonnegative. But if $ba > \pi/2$ and $\cos(ba) \neq 0$, the ODE solution above takes negative values for some x , so it cannot equal $\mathbb{E}_x e^{\gamma\tau}$. The missing hypothesis is integrability of the exponential weight, encoded by the gauge

$$w(x) = \mathbb{E}_x e^{\gamma\tau}.$$

For large positive γ , this gauge is infinite.

Exercise 9.8.C: the gauge and finite-gauge uniqueness

Problem. Let

$$w(x) = \mathbb{E}_x \exp \left(\int_0^\tau c(B_s) ds \right).$$

Assume G is bounded and connected, f, c are bounded and continuous, and $w \not\equiv \infty$. Explain why any bounded solution of

$$\frac{1}{2} \Delta u + cu = 0 \quad \text{in } G, \quad u = f \quad \text{on } \partial G$$

must be

$$u(x) = \mathbb{E}_x \left[f(B_\tau) \exp \left(\int_0^\tau c(B_s) ds \right) \right].$$

Solution. By the gauge theorem in the section, $w \not\equiv \infty$ on a connected bounded domain implies

$$\sup_{x \in G} w(x) < \infty.$$

Let

$$A_t = \int_0^t c(B_s) ds.$$

Exercise 9.8.A shows that $u(B_t)e^{A_t}$ is a local martingale before exit. Stop at $t \wedge \tau$ and use boundedness to get

$$u(x) = \mathbb{E}_x \{ f(B_\tau) e^{A_\tau}; \tau \leq t \} + \mathbb{E}_x \{ u(B_t) e^{A_t}; \tau > t \}.$$

The first term converges by dominated convergence to

$$\mathbb{E}_x \{ f(B_\tau) e^{A_\tau} \},$$

because $|f(B_\tau)e^{A_\tau}| \leq \|f\|_\infty e^{A_\tau}$ and $\mathbb{E}_x e^{A_\tau} = w(x) < \infty$.

For the second term, condition on $\mathcal{F}_t$ and use the Markov property:

$$\mathbb{E}_x(e^{A\tau} \mid \mathcal{F}_t) = e^{At}w(B_t) \quad \text{on } \{\tau > t\}.$$

Since the finite-gauge theorem gives w bounded and bounded away from 0 on G in the way used in Durrett's proof, the tail term is controlled by

$$C \mathbb{E}_x\{e^{A\tau}; \tau > t\} \rightarrow 0$$

again by dominated convergence. Therefore

$$u(x) = \mathbb{E}_x \left[f(B_\tau) \exp \left(\int_0^\tau c(B_s) ds \right) \right],$$

and the bounded solution is unique.

Exercise 9.8.D: the smooth representation solves the PDE

Problem. Let

$$v(x) = \mathbb{E}_x \left[f(B_\tau) \exp \left(\int_0^\tau c(B_s) ds \right) \right].$$

Assume $v \in C^2(G)$. Show that

$$\frac{1}{2}\Delta v + cv = 0 \quad \text{in } G.$$

Solution. Let $B(x, r) \Subset G$ and let

$$\sigma = \inf\{t : B_t \notin B(x, r)\}.$$

By the strong Markov property,

$$v(x) = \mathbb{E}_x \left[e^{\int_0^\sigma c(B_s) ds} v(B_\sigma) \right].$$

For small r , $\mathbb{E}_x \sigma = O(r^2)$ and

$$e^{\int_0^\sigma c(B_s) ds} = 1 + c(x)\sigma + o(r^2)$$

in expectation. Taylor expansion gives

$$\mathbb{E}_x v(B_\sigma) - v(x) = \frac{1}{2}\Delta v(x)\mathbb{E}_x \sigma + o(r^2),$$

because the first moments vanish and the cross terms cancel by symmetry. Combining these expansions,

$$0 = \left(\frac{1}{2}\Delta v(x) + c(x)v(x) \right) \mathbb{E}_x \sigma + o(r^2).$$

Since $\mathbb{E}_x \sigma$ is of order r^2 , divide by $\mathbb{E}_x \sigma$ and let $r \downarrow 0$:

$$\frac{1}{2}\Delta v(x) + c(x)v(x) = 0.$$

Exercise 9.8.E: boundary condition at regular points

Problem. Assume the finite-gauge hypotheses and let y be a regular boundary point. Show that

$$v(x) \rightarrow f(y) \quad \text{as } x \rightarrow y, \ x \in G.$$

Solution. Let $x_n \rightarrow y$ from inside G . Regularity gives, for every $\delta > 0$,

$$\mathbb{P}_{x_n}(\tau \leq \delta) \rightarrow 1, \quad \mathbb{P}_{x_n}(B_\tau \in B(y, \delta)) \rightarrow 1.$$

On $\{\tau \leq \delta, B_\tau \in B(y, \delta)\}$, continuity of f and boundedness of c imply

$$e^{\int_0^\tau c(B_s) ds} f(B_\tau) \rightarrow f(y)$$

as $\delta \downarrow 0$. The part with $\tau > 1$ is negligible because the finite-gauge theorem gives $\sup_G w < \infty$ and

$$\mathbb{E}_{x_n}\{e^{\int_0^\tau c(B_s) ds} |f(B_\tau)|; \tau > 1\} \leq C\|f\|_\infty \mathbb{P}_{x_n}(\tau > 1) \rightarrow 0.$$

The remaining intermediate interval is handled by first taking δ small and then n large. Hence $v(x_n) \rightarrow f(y)$.

Exercise 9.8.F: exponential exit time from an interval

Problem. Let $\tau = \inf\{t : B_t \notin (-a, a)\}$ and $\gamma > 0$. Show that

$$\mathbb{E}_x e^{\gamma\tau} = \frac{\cos(x\sqrt{2\gamma})}{\cos(a\sqrt{2\gamma})} \quad \text{if } 0 < \gamma < \frac{\pi^2}{8a^2},$$

and that $\mathbb{E}_x e^{\gamma\tau} = \infty$ if $\gamma \geq \pi^2/(8a^2)$.

Solution. Let $b = \sqrt{2\gamma}$. If $0 < ab < \pi/2$, then

$$u(x) = \frac{\cos(bx)}{\cos(ba)}$$

is positive on $[-a, a]$ and solves

$$\frac{1}{2}u'' + \gamma u = 0, \quad u(-a) = u(a) = 1.$$

The finite-gauge theorem therefore identifies the Feynman-Kac representation:

$$\mathbb{E}_x e^{\gamma\tau} = u(x) = \frac{\cos(x\sqrt{2\gamma})}{\cos(a\sqrt{2\gamma})}.$$

Now suppose $\gamma \geq \pi^2/(8a^2)$. This is the first Dirichlet eigenvalue threshold for the interval. If the gauge were finite, the finite-gauge theorem would produce a bounded nonnegative solution of

$$\frac{1}{2}u'' + \gamma u = 0, \quad u(\pm a) = 1.$$

But the explicit ODE solution from Exercise 9.8.B either does not exist at the threshold values where $\cos(ba) = 0$, or changes sign when $ab > \pi/2$. Neither possibility can equal the nonnegative function $\mathbb{E}_x e^{\gamma\tau}$. Hence the gauge cannot be finite, and

$$\mathbb{E}_x e^{\gamma\tau} = \infty \quad \text{for all } x \in (-a, a).$$